articulated the synthesis between Mendel’s genetics and Darwin’s theory of natural selection in his book *Evolution: The Modern Synthesis*.

The double helix

Advances in technology such as X-ray crystallography led to more discoveries in the 1940s and ’50s, and the foundation of the new discipline of molecular biology. In 1944, chemist Oswald Avery identified deoxyribonucleic acid (DNA) as the agent for heredity. Rosalind Franklin and Raymond Gosling photographed strands of the DNA molecule in 1952, and James Watson and Francis Crick confirmed its double helix structure the following year. Crick then showed that genetic information is “written” on DNA molecules. The errors that occur when DNA copies itself create mutations—the raw materials for evolution. By the 1980s it was possible to map and manipulate the genes of individuals and species. In the 1990s, the mapping the human genome paved the way for medical research into gene therapy.

Ecologists also want to establish whether genes influence behavior. Back in 1964, William D. Hamilton popularized the concept of genetic relatedness (“kin selection”) to explain altruistic behavior in animals. In *The Selfish Gene* (1976), Richard Dawkins further advanced the gene-centered approach. It is clear that aspects of evolutionary biology will still spark debate as long as ecologists continue to develop Darwin’s theory.

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